Decision support in a smart greenhouse using large language model with retrieval augmented generation

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ABSTRACT

The paper discusses the application of large language models for the development of decision support systems, particularly in the agro-industry. The thesis that decision support is necessary in the sphere of agriculture is confirmed. In order to optimize the activity of smart greenhouses it is reasonable to implement information systems and technologies using large language models. The architecture of decision support system with Retrieval Augmented Generation technology for controlling disease incidence of plants cultivated is proposed. Experiments with the implemented decision support system have been carried out. It is concluded that the application of large language models in the operation of smart greenhouses is promising for use in agriculture in general.

Keywords: smart greenhouse, system architecture, decision support system, Large Language Model, Retrieval Augmented Generation

1. INTRODUCTION

Automated information systems are increasingly used in all areas of human activity. Decision Support Systems (DSS) are complex tools designed to assist decision makers under conditions of uncertainty, complexity or large amounts of data. It can range from simple software applications to complex analytical systems. Such systems can significantly reduce the time to process and solve a problem or issue that has arisen, as well as simplify work and remove some of the routine activities from the person. There are various methods on that an DSS can be based: BI (Business Intelligence), optimization methods, methods of expert systems and knowledge bases, case-based reasoning, machine learning and neural networks, as well as their integration. The most popular approach in the implementation of decision support systems is the use of artificial intelligence. Thanks to it, it became possible to implement and combine speech recognition systems [1], computer vision, text and natural language analysis, including Large Language Models (LLM) [2]. A Large Language Model is an artificial intelligence model it is trained on large amounts of text data and has a large number of parameters [3]. At the moment the use of large language models is becoming more and more widespread. These models trained on huge amounts of textual data are a tool capable of transforming textual information into useful knowledge and making informed decisions. It find applications in a variety of domains, including the development of question-answering systems, chatbots, and decision support systems. One area where the application of LLM can bring significant benefits to humans is agriculture, in particular smart greenhouses [4]. Agriculture is an industry that provides food security and economic prosperity. Optimization of greenhouse processes using modern technologies helps to increase production efficiency, reduce costs and minimize negative impacts. Despite the automation methods in smart greenhouses, that allow optimizing equipment modes, maintaining the right microclimate and nutritional parameters, there are still situations that require specialist intervention. These include, in particular, the emergence of diseases and plant pests, equipment breakdowns, nutritional deficiencies, as well as situations related to operational quality control and business planning. In complex situations, quick and competent decision-making is required, but the experience and knowledge of greenhouse staff is not enough for this. LMM-based IPS is able to act as an expert who has the necessary amount of knowledge and is able to organize a dialogue to analyze the problem, find relevant solutions and explain their conclusions.

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Third International Conference on Digital Technologies, Optics, and Materials Science (DTIEE 2024),
edited by Khamza Eshankulov, Arthur Gibadullin, Proc. of SPIE Vol. 13217, 132170S
© 2024 SPIE · 0277-786X · doi: 10.1117/12.3035606

The aim of the research is to develop a decision support system based on a large language model specifically adapted for use in smart greenhouses. The hypothesis proposed is that the application of a pre-trained LLM tuned to narrow subject knowledge is promising for the development of intelligent DSS. It will help the operators of such greenhouses to find solutions when plant diseases occur and to optimise business processes. The research objectives formulated by the authors include:

- Researching existing LLMs and selecting a suitable one. This task involves reviewing a list of the most popular large language models and analyzing their applicability to the smart greenhouse context. The evaluation is based on factors such as model availability, context window size, number of parameters and the possibility of inference without large computational power.
- Selection of the implementation approach. At this stage, the most appropriate language model and implementation method is selected. This may involve choosing between using a pre-trained model, pre-training it on specific smart greenhouse data, or using state-of-the-art LLM query optimization techniques such as RAG.
- Developing the architecture of the decision making system. At this stage, the system architecture is defined and designed including interaction with the LLM. All components required for the system are also selected.
- Customization and preparation of the selected model. The stage includes preparation of data for the model, quantitation process for faster running of the model, development of instructions and prompts, and tuning of the model for optimal inference.
- Investigating the effectiveness and applicability of the developed system on practical problems. This stage includes testing the system on real data of a smart greenhouse.

2. METHODOLOGY

The authors have carried out a comparative analysis of popular large language models. As a result it was decided to choose a model with the basic architecture Llama2 to implement the decision support system [5]. In addition to the considered characteristics, the choice of this architecture is conditioned by the availability of open-source models, further trained on the Russian-language dataset. As a result, the model "Llama_7b_ru_turbo_alpaca_lora" from the Hugging Face platform is chosen [6]. The presence of the word "lora" in the name of this model means the model was pre-trained using the LoRA training method. LoRA (Low-Rank Adaptation of Large Language Models) is a method for fine-tuning models, that allows only part of the model parameters to be changed. This method activates the gradient control point mechanism in the model. This saves memory during training as only the information required for gradient computation is stored [7].

Although a model with only 7 billion parameters is selected, this has a significant impact on the amount of computational resources required to run the model and hence the speed of model inference. The selected model is still large for local deployment and use. To speed up its operation, it is decided to perform a quantitation process.

Quantitation is a process aimed at reducing the computational and memory overhead when using the model, by converting the model weights and activations to values whose data type has lower precision (e.g. from float16 to int8) [8,9]. The llama.cpp library and the Q5_1 method [10] were used for it. To compare classical language models before and after quantitation we use a quality metric - perplexity, it is defined as the exponential mean negative logarithmic likelihood of all words in the input sequence [10]:

$$PPL(X) = \exp\{-\frac{1}{t}\sum \log p_0(x_i|x_{<i})\} \tag{1}$$

After the quantitation process the model size decreased almost 3 times and the perplexity index for Llama2 models improved from 5.9066 to 5.9481 [9]. In addition, the model is converted to guff format for easy use.

To enable the large language model to generate solutions in the dialogue process and answer questions relevant to the problem situation, context and current conditions at work, a prompt engineering approach with Retrieval Augmented Generation (RAG) is used [11,12]. In this process, a proprietary knowledge base is created from pre-selected documents relevant to the problem domain. The initial query from the user side is converted into a prompt to the model so that the

LMM generates a response based on this knowledge base. The research has used an approach to design the architecture of the SPPR as a system with a software agent that arranges interaction with the LMM and other necessary tools for query generation and dialogue. In this paper, with respect to the problem to be solved, the tools with that the software agent interacts are identified and described. Further, the software implementation of the architecture is performed and experiments with the generation of responses to plant diseases, including with and without the use of RAG are carried out.

3. RESULTS

Before the implementation phase, system design must be carried out. This stage defines the main aspects and architecture of the system to be developed, it helps to speed up the implementation process and improve the efficiency of the system in the future. The general architecture of the system proposed by the authors is presented in Figure 1.

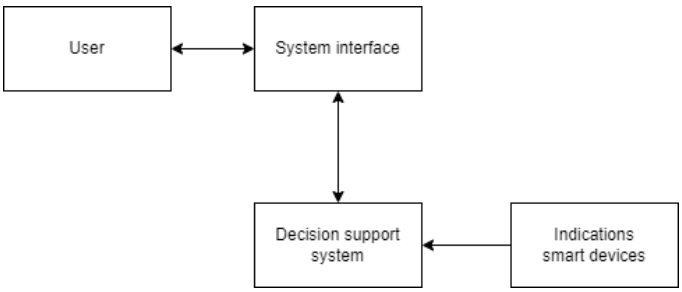

Figure 1. General architecture of the system.

At preset intervals, smart devices send data on air humidity, temperature, as well as data on detection and recognition of diseases cultivated with the use of deep neural networks to the decision support system, within that the obtained data are recorded in a structured database of observations for further use. Based on a LLM the architecture of the decision support system developed by the authors is presented in Figure 2.

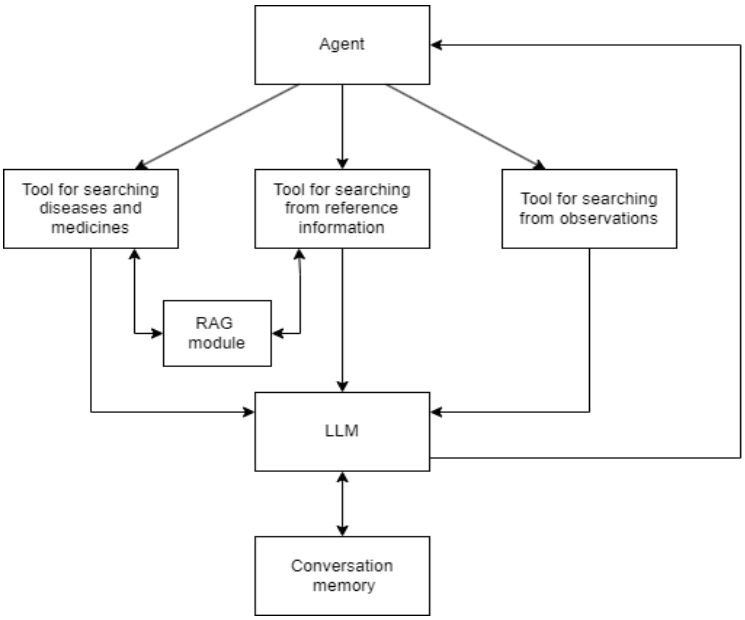

Figure 2. Architecture of the decision support system.

The core of the system is the "Agent" component. It serves as a means to automate and divide tasks by tools, monitors their execution and sends output data to the user. The agent interacts with a large language model, that allows it to understand the user's request in natural language. Through LLM, the agent generates an algorithm to act on the query and a task to execute, determines that tool the task is suitable for and sends it for execution. Tools are the components that the agent uses to solve the tasks. Three tools are defined within this system:

- A tool for searching from observations.
- A tool for searching diseases and medicines.
- A tool for searching from reference information.

The RAG module interacts with the reference search tools and the disease and drug database, as the task of these tools is to respond to user queries with data from downloadable unstructured knowledge bases.

Retrieval Augmented Generation (RAG) is an approach that incorporates a method for extracting relevant information and generating answers based on it. It allows customising the LLM to a specific knowledge domain by embedding the knowledge base in the system architecture [11, 12]. A fragment of the RAG module architecture and interaction with tools for searching from reference information and diseases and medicines is shown in Figure 3.

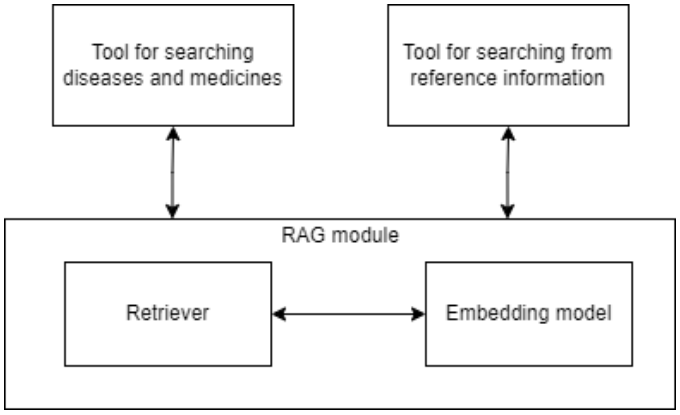

Figure 3. Fragment of RAG module architecture and interaction with tools.

In the developed system RAG module consists of two main components: Embedding model, Retriever. Embedding model is a model with Transformer architecture, the task is to convert a sequence of natural language words into a sequence of embeddings - numerical representations of words [13]. Retriever is a component that is responsible for finding the closest vectors to the vector of query embeddings from the indexed database. An indexed database is a knowledge base that has already undergone the process of converting words into embeddings. It is from it that retriever searches for K most similar vectors to the user's query vector based on vector proximity. The retriever ensemble approach is chosen for the implemented system. In addition to a simple search in vector space, it is decided to use the BM25 search algorithm [14-15].

It should be noted that the tool for searching from observations does not interact with the RAG module, as its task is to search for data about the user-requested observable, the information about that enters the structured knowledge base from the readings of smart devices. The tools within this system consist of a knowledge base (structured or unstructured), a vector database, a prompt, instructions and a linker. The tools within the developed system interact with the LLM agent (Figure 2). Fragments of the architecture of the realized tools are presented in Figure 4. A knowledge base is a set of text documents, books or other resources containing information on a particular subject area or topic.

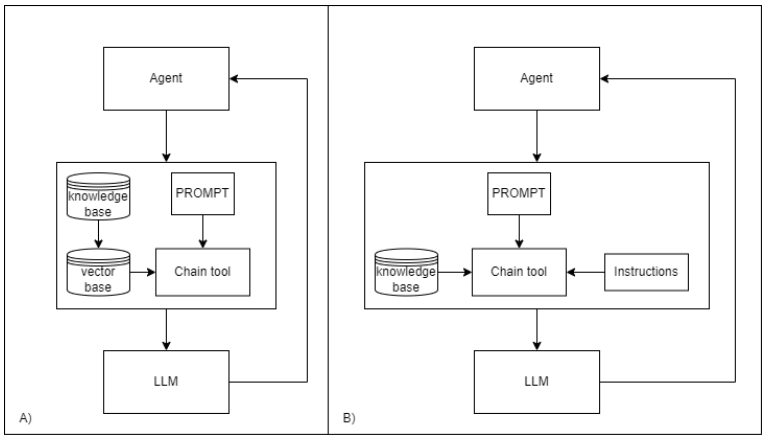

Figure 4. Fragment of tool architecture interacting with LLM. A) Fragment of the architecture of tools interacting with the RAG module. B) Fragment of the tool architecture for searching from observations.

The system under development includes three knowledge bases:

- The database of observations and readings of smart devices is a structured data set that describes various indicators of the observed plants and their environment. The database is systematically populated and updated with data from the smart greenhouse devices. It is used by the tool to search from observations.

- Diseases and medicines database - represents a reference book containing information about various plant diseases and their pests, as well as treatment methods, descriptions of medicines, dosages and methods of application. This database is used as a tool to search for diseases and medicines.

- Reference knowledge base - a database includes all other data useful for the smart greenhouse. For example, it may contain tips on how to care for or grow plants, etc. This base is used by the search tool from the help information. All the above bases are depicted as "Knowledge base" in Figure 4.

Vector database is a system for storing, processing and outputting information with data represented as vectors. The system under development contains two vector bases, one of that is part of the disease and drug search tool and the other is part of the search tool from reference information. It stores data from knowledge bases that were previously chunked and converted into vectors using Embedding model (Figure 3). This approach allows to quickly search for answers to queries and rank them by vector similarity.

One of the important aspects when using LLM is PROMPT. Prompt is a key element in working with LLM. It is the short text or instructions that are provided to the model to generate a response. The PROMPT should define what is to be obtained from the model and should aim to help the model understand the context of the request. The more clear and precise the PROMPT is, the more relevant the output of the model will be. The PROMPT can specify the model's role and responsibilities, as well as the location or subject area. A good technique is to emphasise the responsibility and importance of the model's work. This results in improved response generation and a reduction in errors or "hallucinations" of the model. An example of one of the PROMPTs used in the system being implemented is shown in Figure 5.

```
“You are an assistant on a smart farm. Your main task is to help farmers in taking care of plants.
You analyze the available data to identify plant health indicators and assist in case of diseases.
Your responsibilities include analyzing incoming data, detecting anomalies, diagnosing possible plant diseases,
and providing treatment recommendations. You also answer farmers' questions about the condition of plants
 and ways to care for them. It is important to remember that decision-making data should be taken
 from the context: {context}. If you don't know the answer, there's no need to invent a false one;
it's important to honestly say that you don't know the answer to the given question,
as the quality of your answers affects crop levels and human life. Be attentive!”
```

Figure 5. Example of the PROMPT in use.

4. DISCUSSION

Thanks to the agent, the queries coming from the user are distributed and executed by the appropriate tools. This approach improves the performance of the RAG module, as the knowledge bases are divided by type: drugs and diseases, reference information, etc. This prevents inappropriate information for the question being answered. This prevents information that is inappropriate for the question being asked from being included in the answer.

The database of observations and readings of smart devices (Figure 4) has the following indicators: id of the plant, date and time of fixation of readings, air humidity, air temperature, soil type of the plant, presence of disease, probability of disease, name of the disease and others. This structure allows information about a particular plant to be requested from the system at a specific time, thus optimizing the observation process and speeding up information retrieval.

One of the algorithms for using the implemented system can be represented as follows:

- The user can request a list of diseased plants and the system, having analyses a structured knowledge base updated with readings from smart devices, will provide their identifiers.
- The user can enquire about the status or presence of disease in a particular plant and get a response from the system.
- Next, information about the disease can be requested. Given the context of the dialogue, a search from the knowledge base of diseases and drugs will be performed using the RAG module, then an answer describing the disease will be given.
- The user can also request a treatment method for the disease, and this request is also searched using RAG from the disease and drug knowledge base.

An example of interaction with the system according to the above algorithm is shown in Figure 6.

```
Q: ID of plants that are sick?
--------------------------------------------------------------------------------
A: Plants with id 2, 3 and 8 have the disease.
--------------------------------------------------------------------------------
Q: Name of the disease in the plant with id 2
--------------------------------------------------------------------------------
A: Disease - Blackleg.
--------------------------------------------------------------------------------
Q: Tell me about this disease
--------------------------------------------------------------------------------
A: Most often, Blackleg is found in seedlings of tomatoes, peppers, eggplants, cucumbers,
cabbage, radishes, lettuce, as well as petunias and other flowers.
The fight against blackleg is primarily based on the prevention of this disease.
Buy high-quality plant seeds and pay attention to disease-resistant varieties.
--------------------------------------------------------------------------------
Q: How to treat this disease in tomatoes?
--------------------------------------------------------------------------------
A: First you need to remove all affected seedlings. The remaining plants
 should be watered at the root with a solution of Fitosporin
 (100 ml of the drug per 10 liters of water) and the seedlings should be sprayed with it.
--------------------------------------------------------------------------------
```

Figure 6. Example of interaction with the developed decision-making system.

The implemented system is also capable of processing complex queries by breaking them down into a number of simpler questions. The received questions are sequentially analysed, the answer to each of them is generated and then the overall final result is given. An example of a response to a complex query is shown in Figure 7.

```
Q: Analyze the plant with id 8. What is wrong with it and how to deal with it?
--------------------------------------------------------------------------------
A: Plant with id 8 is potato. The air temperature is +27°C, air humidity is 87%.
The plant has a disease - Macrosporiosis. Macrosporium (dry spotting) is a disease
caused by imperfect fungi MacrosporiumsolaniEllisetMartin.
Plants from the nightshade family are most susceptible to disease.
The pathogen spreads through the air through droplets of moisture.
The pathogen develops in a humid environment at a temperature
of +20... +27°C; only the anamorphic stage of development is noted in the cycle.
Suitable medicine: Copper sulfate. Directions for use:
dilute 100g of the drug in 10 liters of water, spray the plant and  let it dry.
Repeat the process three times.
For prevention, the air humidity should be reduced to 70-75%.
--------------------------------------------------------------------------------
```

Figure 7. An example of a typical system response to a complex query.

The main weakness of large language models is their "hallucinations". This effect manifests itself in the fact that the model, not knowing the answer to a question, starts to generate a false answer. Obviously, this can have a negative impact on decision making. Thanks to the RAG module, the given prompt and instructions, the model integrated in the system does not generate a false answer at the moment when it does not receive information in the context, but issues a warning that it is unable to answer the question posed. This approach reduces various risks, helps to explicitly find gaps in the knowledge base and fill them in a timely manner. An example of a query that the system was unable to answer is shown in Figure 8.

```
Q: Name of the disease in the plant with id 3?
----------------------------------------------------------------
A: Powdery mildew.
----------------------------------------------------------------
Q: How to treat this disease?
----------------------------------------------------------------
A: Unfortunately I have no missing information about this.
----------------------------------------------------------------
```

Figure 8. An example of a query the system was unable to provide an answer.

The results of this study allow us to positively assess the prospects of LLM and RAG for creating decision support systems that will work like an experienced subject matter expert - recognizing a complex situation, proposing solution options for this situation and answering additional questions from the user, including explaining the proposed solutions. Today in Agriculture much attention is paid to the application of AI and, in particular, neural networks for automatic recognition of plant diseases, pests or classification of agricultural products [16-20]. The proposed approach allows to complement these tools with new capabilities. Now, after detecting a problem situation with the help of agents working with LMM, RAG and other tools, it will be possible not only to detect and recognize a problem situation, but also to propose a reasoned expert solution for it.

5. CONCLUSION

In this paper, the decision support problem of operating smart greenhouses using LLM is considered. An analysis of currently popular large language models is presented and a suitable model for implementation is selected. The process of training the model by quantizing it to speed up the inference is described. The overall architecture of the decision-making system and its individual blocks is designed and discussed, and an example of the use of a decision-making system adapted for a smart greenhouse is given. It is shown the use of LLM and RAG allows finding solutions in the knowledge base when complex situations arise and offering these solutions in dialogue with the user. At the same time, the own knowledge base allows to move from general recommendations, that the pre-trained language models are

capable of, to specific recipes and action plans adequate to the available conditions of the smart greenhouse. In experiments the adapted and quantized model demonstrates responsiveness to a variety of tasks and can be deployed on agro-enterprise servers or in the cloud. This confirms the accessibility of this approach for building applied DSS in agroculture. Thus, the conducted research supports the hypothesis that the application of LLM in smart greenhouses is promising for the optimisation and development of agriculture. Further research in this area can focus on improving the algorithms and integrating the developed decision support systems into real-world smart greenhouse environments. Such improvements will help to complement the automation suite of smart greenhouses, urban farms and other agro-industrial enterprises with expert decision support systems for specialists in complex situations.

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